

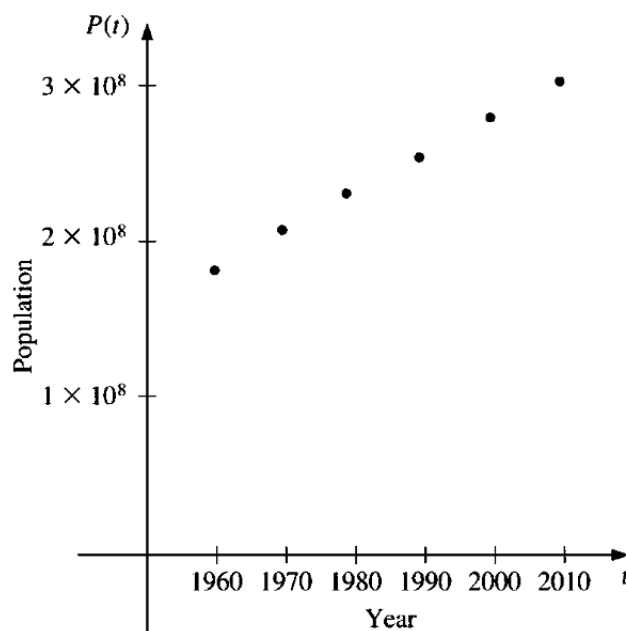
Section 1: Foundations of Interpolation

Here we focus on data that depends on one independent variable.

e.g. $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

Consider US census data, every 10 years from 1960.

Year	1960	1970	1980	1990	2000	2010
Population (in thousands)	179,323	203,302	226,542	249,633	281,422	308,746



How can we estimate the population in 1998?

Polynomial Interpolation

A polynomial in the standard form is $P_n(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ where n is a non negative integer and $a_0, \cdots, a_n$ are real constants. Polynomials are widely used in interpolation because they allow for easy application of fundamental mathematical operations such as differentiation and integration.

1.1: Lagrange Interpolation

Case 1: Linear Polynomial Interpolation

Find a linear polynomial interpolating a function $f(x)$ at (x_0, y_0) and (x_1, y_1) where $f(x_0) = y_0, f(x_1) = y_1$.

Case 2: Quadratic Polynomial Interpolation

Consider the construction of a polynomial of degree at most 2 that passes through three points $(x_0, y_0), (x_1, y_1), (x_2, y_2)$.

Example 1. Find the linear Lagrange polynomial passing through the points $(0, 3), (1, 0), (2, 2)$.

Case 3: General Case

Theorem 1

If $x_0, \dots, x_n$ are $n+1$ distinct numbers and $y_0, \dots, y_n$ are the corresponding values, then a unique polynomial $P_n(x)$ of degree at most n exists with

$$P_n(x_i) = y_i, \quad i = 0, 1, \dots, n.$$

This polynomial is given by

$$P_n(x) = \sum_{i=0}^n y_i l_i(x)$$

where

$$l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

$$l_i(x_j) = \delta_{ij} \quad 0 \leq i, j \leq n.$$